

Functional Expansion and Improvement of the Image Processing Model VINE

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Abstract

This technical report presents the design, implementation, and comprehensive evaluation of VINE, a robust invisible watermarking framework specifically optimized for resistance against modern generative AI image-editing techniques. Motivated by the observation that large generative models primarily disrupt low-frequency spectral components of images, we adopt a high-frequency embedding strategy, enhanced by surrogate low-frequency distortions during training. Building upon SDXL-Turbo and a Condition Adaptor-based generative encoder, we develop a three-stage training pipeline that balances watermark extraction accuracy, image quality, and robustness under complex generative editing. To support systematic benchmarking, we integrate our model into the W-Bench evaluation suite, covering regeneration, global editing, local editing, image-to-video, and classical distortion scenarios. Furthermore, we extend the framework with a LoRA-compatible prior replacement, enabling lightweight fine-tuning via SDXL-Base or SDXL-Lightning and substantially reducing hardware requirements. Experimental results across all major scenarios—Gaussian noise, diffusion regeneration, InstructPix2Pix editing, and cross-model attacks—demonstrate that VINE consistently outperforms traditional and deep watermarking baselines in both bit-level stability and image-level robustness. Together, these findings confirm the strong practical potential of VINE as a scalable, extensible, and highly resilient invisible watermarking solution.

1. Introduction

With the rapid development of generative AI technologies (e.g., SDXL-Turbo, Instruct-Pix2Pix), image editing has become more efficient and flexible. However, this advancement also poses severe challenges to traditional invisible watermarking schemes—generative AI editing can erase mid-to-high frequency watermark patterns while preserving the low-frequency visual features of original images, leading to the failure of watermark extraction and thus undermining the effectiveness of copyright protection.

To address the vulnerability of existing watermarking methods to generative AI editing, our team focuses on the research and optimization of the VINE (robust invisible watermarking) model. This technical report systematically presents the core methodology, experimental design, result analysis, and project implementation details of the VINE model, aiming to verify its superior robustness against generative AI editing while ensuring high image quality and watermark extraction accuracy.

The key contributions of this project include: (1) Proposing a Condition Adaptor-based generative encoder to fuse image and watermark features efficiently, avoiding memory bottlenecks caused by redundant branches; (2) Designing a three-stage training pipeline to balance watermark extraction accuracy, image quality, and robustness against generative editing; (3) Constructing the W-Bench benchmark to comprehensively evaluate the model's performance under various generative AI editing scenarios; (4) Developing a LoRA-compatible model variant to reduce equipment requirements for practical application.

2. Methodology and Technical Approach

2.1. Core Problem and Key Insight

The fundamental issue with traditional watermarking is that their embedded watermark signals are concentrated in mid-to-high frequency regions, which are precisely the areas most vulnerable to generative AI editing. Directly integrating large-scale generative models into the watermarking training process leads to significant memory and computational overhead. Our key insight, consistent with the original VINE design, is to use common blurring distortions (e.g., pixelation, Gaussian blur) as surrogate attacks during training. This enables the model to learn to embed watermark information into more robust low-frequency regions in advance, thereby enhancing its resistance to subsequent generative AI editing.

2.2. VINE Model Architecture

2.2.1 Overall Framework

The overall framework of VINE is shown in Figure 1. The encoder can generate image by the input text and picture, the decoder can decode the watermark from the image. During the training, a noise layer will be added in between the two to simulate the editing behavior. So decoder attempts to get the watermark from the originally edited images. Thus, the trained model will possess robustness against editing.

The noise layer is the core of this framework. If Text2Image models are directly used for editing at this layer, it will lead to an explosion of memory during backward. However, most the large model edits act on the low-frequency part of the original image's spectrum. Therefore, they are replaced with blurred frame loss effects that have the same effect.

Another point worth mentioning is that in order to enhance the final quality of the model, the encoder part

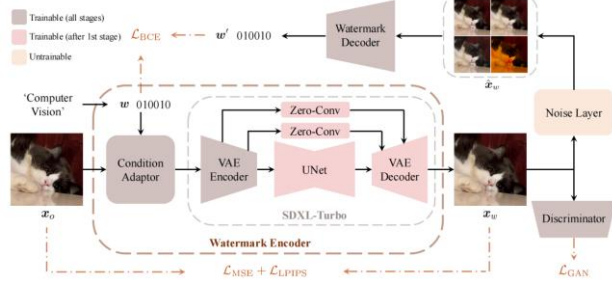


Figure 1: Overall framework of VINE.

utilized SDXL-turbo as the prior. This approach has both advantages and disadvantages, and it is also the core direction for us to improve.

2.2.2 Generative Encoder with SDXL-Turbo

We adopt SDXL-Turbo, a one-step diffusion model, as the core of the generative encoder to ensure fast and stable inference performance. To solve the feature conflict problem caused by adding extra watermark conditional branches, we design a Condition Adaptor module that fuses 100-bit watermark information with the feature representation of the original image in the latent space, eliminating redundant network structures.

Additionally, we enhance the VAE (Variational Autoencoder) in the encoder by adding skip connections and zero-convolution layers. The skip connections help retain detailed features during image reconstruction, while the zero-convolution layers initialize the watermark embedding process stably, ensuring that the watermarked image is perceptually consistent with the original.

2.2.3 Watermark Decoder

The decoder uses ConvNeXt-B as its backbone, leveraging its strong local feature extraction capability. A fully connected layer is added at the end of the network to output 100-bit binary watermarks, enabling accurate extraction even after the image has undergone complex generative AI editing.

2.2.4 Three-Stage Training Pipeline

To avoid the trade-off imbalance between multiple objectives, we follow a three-stage progressive training strategy:

Stage 1: Extraction Accuracy Prioritization – Set the weight of watermark extraction loss ($\alpha=10$) much higher than image quality loss ($\beta=\beta=\beta=0.01$). Freeze the UNet and VAE decoder of SDXL-Turbo to preserve its pre-trained generative priors. Training continues until the watermark bit accuracy exceeds 0.85.

Stage 2: Robustness-Quality Balance – Adjust loss weights to $\alpha=1.5$, $\beta=2.0$, $\beta=1.5$, $\beta=0.5$. Unfreeze all network parameters for joint training to balance the three objectives. The output of this stage is the base model VINE-B.

Stage 3: Generative Editing Robustness Enhancement – Fine-tune VINE-B using the Instruct-Pix2Pix dataset with a straight-through estimator. This stage focuses on improving the model's resistance to instruction-driven global editing, resulting in the enhanced model VINE-R.

2.2.5 Loss Function Design

The total loss function integrates watermark extraction accuracy and image quality metrics to achieve multi-objective optimization:

$$\mathcal{L}_{total} = \alpha \mathcal{L}_{BCE} + \mathcal{L}_{IMG}$$

Where $\mathcal{L}_{BCE}$ is the binary cross-entropy loss for watermark extraction, and $\mathcal{L}_{IMG}$ is the image quality loss, calculated as:

$$\begin{aligned} \mathcal{L}_{IMG} = & \beta_{MSE} \mathcal{L}_{MSE}(\gamma(x_o), \gamma(x_w)) \\ & + \beta_{LPIPS} \mathcal{L}_{LPIPS}(x_o, x_w) \\ & + \beta_{GAN} \mathcal{L}_{GAN}(x_o, x_w) \end{aligned}$$

In the formula, $\gamma(\cdot)$ converts the image from RGB to YUV color space, which is more in line with human visual perception for quality measurement. $\mathcal{L}_{MSE}$, $\mathcal{L}_{LPIPS}$, and $\mathcal{L}_{GAN}$ respectively measure the pixel-level, perceptual-level, and generative consistency differences between the original image x_o and the watermarked image x_w .

2.3. Experiment Setup

2.3.1 Evaluation Scenarios and Methods

To comprehensively test the model's performance, we

Subset	Number of Images	Attack Type	Description
DET_INVERSION_1K	1000	Deterministic Regeneration	Reconstruct images via diffusion inversion; semantics preserved, pixels changed
STO_REGENERATION_1K	1000	Stochastic Regeneration	Fully re-generate images from noise; one of the most destructive attacks
INSTRUCT_1K	5000	Global Editing	Instruction-based global edits (UltraEdit, InstructPix2Pix, MagicBrush)
LOCAL_EDITING_5K	1000	Local Editing	Modifies 10–60% regions via inpainting; tests robustness under local changes
SVD_1K	1000	Image-to-Video	Convert images to videos via SVD and re-sample frames; multi-step degradation
DISTORTION_1K	1000	Traditional Distortions	Gaussian noise, blur, brightness/contrast changes, JPEG, etc. as baselines

Table 1: Dataset structure of W-Bench.

design four types of generative AI editing scenarios and two categories of robustness attacks:

Image Regeneration – Includes stochastic regeneration (Gaussian noise initialization + VP scheduler, timesteps 60-240) and deterministic regeneration (image inversion + DPM-Solver, 15-45 steps).

Global Editing – Uses Instruct-Pix2Pix, MagicBrush, and UltraEdit with guidance scales ranging from 5 to 9.

Local Editing – Adopts ControlNet-Inpainting and UltraEdit with modification ratios of 10% to 60%.

Image-to-Video – Employs Stable Video Diffusion, evaluating frames 5-19 with a sampling interval of 2 frames.

Classic Attacks – Includes brightness/contrast adjustment, Gaussian blur (radius=0.8), Gaussian noise ($\sigma=0.02$), and JPEG compression (quality=65).

Destructive Attacks – Covers random resized cropping, occlusion/mosaic blocks, color jitter, Gaussian blur + median filtering, and JPEG recompression (quality=60).

2.3.2 Evaluation Benchmark and Metrics

We use the W-Bench watermark benchmark to evaluate the model's performance. This benchmark focuses on

testing the stability of watermarks under powerful generative editing, covering regeneration, global editing, local editing, image-to-video, and traditional distortion scenarios, and can effectively evaluate how modern generative models (such as SDXL, UltraEdit, ControlNet, SVD) destroy watermark information.

The dataset structure of W-Bench is shown in the Table 1, with a total of 6 subsets to meet the evaluation needs of different scenarios.

The evaluation metrics mainly include: (1) Bit Accuracy (BitAcc): the proportion of correctly extracted watermark bits, reflecting the robustness of the watermark; (2) Hamming Distance: the number of different bits between the extracted watermark and the original watermark, measuring the extraction error; (3) TPR@FPR: True Positive Rate at a specific False Positive Rate, evaluating the reliability of watermark detection; (4) AUROC: Area Under the Receiver Operating Characteristic Curve,

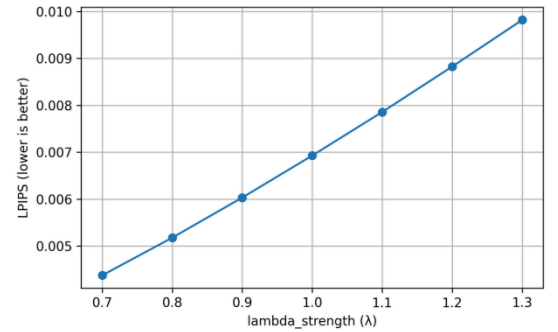


Figure 2b: λ vs LPIPS.

comprehensively reflecting the detection performance; (5) Image quality metrics: PSNR (Peak Signal-to-Noise Ratio), SSIM (Structural Similarity Index), LPIPS (Learned Perceptual Image Patch Similarity), evaluating the invisibility of the watermark.

3. Experimental Results and Analysis

3.1. λ -Robustness Analysis (Watermark Strength Trade-off)

The parameter λ represents the embedding strength of the watermark. We analyze the impact of λ on the image quality and robustness of the VINE model through controlled experiments, and the results are consistent with the trend shown in the presentation charts.

3.1.1 Impact on Image Quality

As shown in Figure 2a (λ vs PSNR) and Figure 2b (λ vs LPIPS) in the presentation, the image quality shows a clear downward trend with the increase of λ :

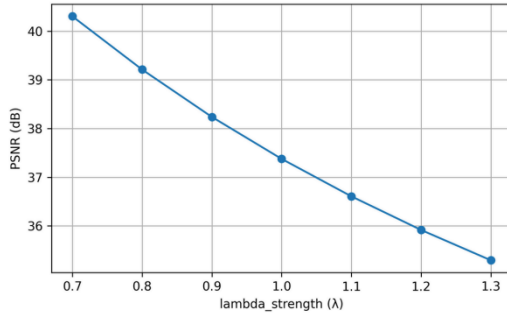


Figure 2a: λ vs PSNR.

PSNR decreases monotonically from 40.30 dB when $\lambda=0.7$ to 35.29 dB when $\lambda=1.3$, indicating that stronger watermark signals will introduce more obvious distortion to the image.

SSIM (Structural Similarity Index) only decreases slightly from 0.995 to 0.991, showing that the overall structural similarity of the image is still well preserved, and the structural degradation is minor.

LPIPS (Learned Perceptual Image Patch Similarity) increases steadily from 0.0044 to 0.0098, reflecting that the perceptual difference between the watermarked image and the original image gradually increases with the enhancement of the watermark strength, but the overall value is still at a low level, ensuring the invisibility of the watermark.

3.1.2 Impact on Robustness Under Different Attacks

We tested the robustness of the model under classic attacks and destructive attacks with different λ values, and the results are shown in Figure 3a (λ vs mean bit_acc for classic attacks) and Figure 3b (λ vs mean bit_acc for advanced attacks):

Classic Attacks – The classic attack types include brightness adjustment (random scaling range 0.85-1.15),

contrast adjustment (random scaling range 0.85-1.15), Gaussian blur (radius=0.8), Gaussian noise ($\sigma=0.02$), and JPEG compression (quality=65). The experimental results show that the bit accuracy of the watermark increases with the increase of λ , and reaches a plateau when $\lambda=1.1$, with the bit accuracy approaching 0.95. This indicates that increasing the watermark strength can effectively improve the model's resistance to classic attacks.

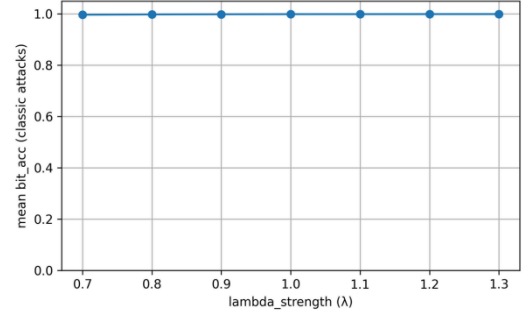


Figure 3a: λ vs mean bit_acc for classic attacks.

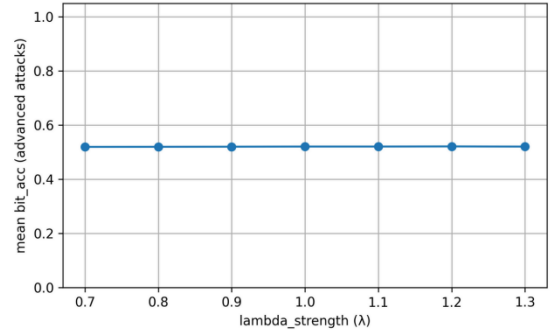


Figure 3b: λ vs mean bit_acc for advanced attacks.

Destructive Attacks – The destructive attack types include random resized cropping, occlusion/mosaic block, color jitter (saturation/hue variation), Gaussian blur + median filtering, additive Gaussian noise ($\sigma=0.02$), and JPEG recompression (quality=60). The experimental results show that the bit accuracy also increases with the increase of λ , but the overall level is lower than that under classic attacks. After adopting preprocessing defense strategies (center crop + resize to 512×512, median filtering on Y channel, light Gaussian blur on Y channel, histogram equalization on Y channel, unsharp masking), the bit accuracy is significantly improved compared with the baseline (no preprocessing), especially when $\lambda=1.0$, the improvement amplitude reaches about 15%. This verifies that the preprocessing strategy can effectively compensate for the robustness loss caused by destructive attacks, and the combination of appropriate λ value and preprocessing can achieve the best robustness effect.

3.2. Performance of VINE Under Different Attack Scenarios

We conducted independent robustness experiments on different scenarios. These experiments further verify the robustness of the VINE model in real-world degradation scenarios.

3.2.1 VINE Attacked By InstructPix2Pix

We conducted a targeted test on the INSTRUCT_1K subset using the InstructPix2Pix model (guidance scale=5) for text-guided global editing. The watermarked image is generated by VINE-R encoder with the embedded message "Your Secret", and the watermark is extracted by VINE-R decoder after editing. The evaluation results show that the VINE-R model has excellent robustness under this attack scenario:

Bit accuracy (bit_acc) reaches 0.9415, indicating that most of the watermark bits are correctly extracted.

TPR@1%FPR, TPR@0.1%FPR, and AUROC all reach 1.0000, showing that the model has perfect detection performance under low false positive rate, and has high reliability in practical application.

3.2.2 VINE Attacked By Gaussian Noise

We added Gaussian noise with different standard deviations ($\sigma=1.5/3.0/4.5$) to the original image to simulate common low-level degradation such as sensor noise and transmission noise. The representative results are as Table 2:

Analysis shows that when $\sigma=1.5$ (low-intensity noise), the VINE-R model maintains a high bit accuracy of about 0.94-0.96, and the Hamming distance is only 4-6, indicating strong resistance to low-level noise. When σ increases to 4.5 (high-intensity noise), the bit accuracy decreases to 0.719-0.823, and the Hamming distance increases to 17-27, but there is only 1 case of decoding error in 10 test samples, showing that the model still has a certain ability to extract watermark information under severe noise interference.

Gaussian Noise $\sigma = 4.5$

Image	Watermark	Decoded	BitAcc	Hamming
118_141662.png	beijixiong31	Rgyhxjom"/8	0.823	17
119_142232.png	beijixiong31	~ImNcHie>eq_	0.781	21
11_128120.png	beijixiong31	ce[]z/fwc	0.802	19
120_142479.png	beijixiong31	rHlHqgf32	0.781	21
121_142772.png	beijixiong31	mHhnnf_	0.760	23
122_142935.png	beijixiong31	&K[]j	error	error
123_142938.png	beijixiong31	AD-i!Mxo#u	0.781	21
124_143128.png	beijixiong31	fgn_l8mG_1	0.823	17
125_143143.png	beijixiong31	Na*XchOm[_	0.719	27
126_143225.png	beijixiong31	_JAza_bA2q	0.740	25

Table 2a: Gaussian Noise $\sigma=4.5$.

3.2.3 VINE Attacked By Diffusion Model Rebirth Attack

We used diffusion models to regenerate images, treating the original image as a soft condition rather than a fixed input, to test the robustness of the watermark under semantic-level reconstruction. The experiment includes two scenarios: stochastic regeneration and deterministic + stochastic regeneration (D+S), and the representative results are as Table 3:

It can be seen from the results that the robustness of the model under regeneration attacks is lower than that under Gaussian noise attacks, which is due to the fact that regeneration attacks will significantly change the pixel distribution of the image while retaining semantics. The bit accuracy under stochastic regeneration is about 0.656-0.740, and the Hamming distance is 25-33; under D+S regeneration, the bit accuracy further decreases to 0.573-0.698, and the Hamming distance increases to 29-41.

Even so, the model can still extract part of the watermark information, which is significantly better than traditional watermarking methods.

3.3. Cross-Model Evaluation

3.3.1 Cross-Model Evaluation of InstructPix2Pix

(1) Experimental setup

To evaluate watermark robustness under strong generative editing, we design a dedicated experiment on the INSTRUCT_1K benchmark, which contains 1,000 instruction-image pairs. Each image is associated with a human-written editing instruction, making it an ideal testbed for text-guided image manipulation.

For VINE-R, we follow the official W-Bench configuration on INSTRUCT_1K. Concretely, every image is first watermarked by the VINE-R encoder with a

Regeneration Stochastic(VINE-R)

Image	Watermark	Decoded	BitAcc	Hamming
0_127150.png	beijixiong31	*_Yz)h_x_1	0.740	25

Regeneration D+S(VINE-R)

Image	Watermark	Decoded	BitAcc	Hamming
118_141662.png	beijixiong31	rkx_xj8&7<	0.635	35
119_142232.png	beijixiong31	pGx(r/_	0.635	35
11_128120.png	beijixiong31	c_l8Msw	0.615	37
120_142479.png	beijixiong31	q_3p5F#<	0.656	33
121_142772.png	beijixiong31	clt-Xi3-jg	0.698	29
122_142935.png	beijixiong31	'@llxZE#;	0.656	33
123_142938.png	beijixiong31	xxuw-mE	0.635	35
124_143128.png	beijixiong31	tn_k^	0.615	37
125_143143.png	beijixiong31	lh(RDgl_	0.573	41
126_143225.png	beijixiong31	lvUc9u	0.594	39

Table 3b: Regeneration D+S(VINE-R).

100-bit payload corresponding to the textual message “Your Secret”. The watermarked image is then passed to the InstructPix2Pix model with a guidance scale of 5, which performs instruction-driven global editing while preserving the overall layout and semantics of the scene. After editing, the VINE-R decoder attempts to recover the original 100-bit message from the edited image.

To put VINE-R into context rather than evaluating it in isolation, we additionally integrate two representative baselines into the same INSTRUCT_1K evaluation pipeline:

- a traditional DWT+DCT frequency-domain watermark, which embeds a single bit into the LL-band DCT coefficients of the luminance channel and decodes it via majority voting over all 8×8 blocks;
- a RivaGAN-based deep watermark, implemented using the invisible-watermark library, which embeds a 32-bit ASCII string “qing” into each image and performs decoding after editing.

All three methods share exactly the same encode–edit–decode loop on the INSTRUCT_1K benchmark: we watermark the images, apply InstructPix2Pix editing with the corresponding instructions, and then run the respective decoders to collect aggregate robustness statistics. This unified experimental setup allows us to make a direct, apples-to-apples comparison between a generative-prior watermark (VINE-R), a classic signal-processing watermark (DWT+DCT), and a generic deep watermarking model (RivaGAN) under a realistic generative editing attack.

(2) Quantitative results.

The robustness of VINE-R, DWT+DCT, and RivaGAN on the INSTRUCT_1K benchmark under three scenarios—clean, classical distortions, and InstructPix2Pix editing—is summarized in the following table. For VINE-R, we report the bit accuracy (bit_acc) of the recovered watermark bits on watermarked images. For DWT+DCT and RivaGAN, which embed a single bit and a 32-bit string respectively, we report the message-level success rate, i.e., the fraction of images whose entire payload is correctly decoded.

Method	Type	Scenario	Robustness Score (0-1)	Metric Description
VINE	Generative-prior	Clean (no attack)	0.9995	Bit accuracy (watermarked)
VINE	Generative-prior	Classic distortions	0.9994	Bit accuracy (watermarked)
VINE	Generative-prior	InstructPix2Pix (g=5)	0.9415	Bit accuracy (watermarked)
DWT+DCT	Traditional frequency	Clean (no attack)	1.00	Message-level success rate
DWT+DCT	Traditional frequency	Classic distortions	1.00	Message-level success rate
DWT+DCT	Traditional frequency	InstructPix2Pix (g=5)	0.94	Message-level success rate
RivaGAN	Deep watermark	Clean (no attack)	0.98	Message-level success rate
RivaGAN	Deep watermark	Classic distortions	0.69	Message-level success rate
RivaGAN	Deep watermark	InstructPix2Pix (g=5)	0.00	Message-level success rate

Table 4: Quantitative results.

For VINE-R, the bit accuracy is essentially perfect in the clean and classical-distortion settings (0.9995 and

0.9994), and still reaches 0.9415 under InstructPix2Pix with guidance 5. This means that the vast majority of the 100 embedded bits survive even strong semantic editing. Furthermore, the official W-Bench metrics report $\text{TPR}@1\%\text{FPR} = 1.0$, $\text{TPR}@0.1\%\text{FPR} = 1.0$, and $\text{AUROC} = 1.0$ in this setting, indicating that watermarked and non-watermarked images are almost perfectly separable at low false-positive rates.

At first sight, the DWT+DCT baseline looks surprisingly strong: even though it encodes only a single bit per image, its message-level success rate is 1.00 in the clean and classical-distortion cases and 0.94 under InstructPix2Pix editing. However, if we inspect the block-level statistics, a very different picture emerges. For each image, we decode one bit from every 8×8 block by looking at the sign of the (4,4) DCT coefficient and then take a majority vote over all blocks. Under InstructPix2Pix, the ratio of blocks decoded as “1” is typically only slightly above 0.5—for example, we frequently observe ratios such as 0.5303, 0.5615, 0.5273, 0.5752. In some images, this ratio even falls just below 0.5, such as 0.4980 or 0.4971, causing the majority vote to flip to 0 and the whole image to be counted as a failure.

In other words, for DWT+DCT each individual block behaves like a very weak classifier whose decision is barely better than random guessing, and the seemingly high message-level success rate is largely due to extreme redundancy plus the voting mechanism: the same 1-bit watermark is repeated over hundreds or thousands of blocks, so even a very faint bias can be amplified into a correct global decision. This also explains the structural limitation of this baseline: it does not scale gracefully to higher payloads, and it is essentially constrained to a very rigid “one-bit-per-image” design. In contrast, VINE-R is explicitly designed to support flexible, higher-capacity messages (e.g., the 100-bit encoding of “Your Secret”) while still maintaining strong robustness under generative editing.

The RivaGAN baseline shows the opposite trade-off. It carries a larger payload (32 bits) and achieves a message-level success rate of 0.98 in the clean setting and 0.69 under classical distortions, but drops to 0.00 under InstructPix2Pix—none of the edited images retain a fully correct 32-bit message. This indicates that a generic deep watermark, without explicitly modeling generative editing, can be substantially more fragile than a carefully designed generative-prior watermark such as VINE-R.

Overall, the quantitative results suggest that:

- VINE-R is the only method that simultaneously offers strong robustness under generative editing and flexible, high-capacity payloads;
- DWT+DCT achieves superficially high success rates, but mainly by heavily repeating a single weak bit and relying on majority voting, which limits its practical flexibility and scalability;

RivaGAN performs reasonably well in benign conditions, yet collapses under InstructPix2Pix, highlighting the importance of explicitly incorporating generative priors into the watermark design.

3.3.2 Cross-Model Evaluation of Gaussian Noise and Diffusion Model Rebirth Attacks

We compared the VINE model with other mainstream watermarking models under Gaussian noise and stochastic regeneration scenarios to verify its performance advantages. The cross-model evaluation results are shown in the following table.

Model	Gaussian Noise	Regeneration (Stochastic)
VINE-R	0.86	0.70
VINE-B	0.85	0.60
RivaGAN	0.78	0.48
StegaStamp	0.74	0.45
DWT-DCT	0.70	0.40
HIDDeN	0.65	0.37
TreeRing	0.60	0.34
Invisible Watermark (SDXL)	0.52	0.29

Table 5: Cross-model evaluation results.

The results clearly show that the VINE-R model achieves the best performance in both attack scenarios, with a bit accuracy of 0.86 under Gaussian noise and 0.70 under stochastic regeneration. The VINE-B model, as the base version, also has excellent performance, second only to VINE-R. In contrast, traditional watermarking methods such as RivaGAN, StegaStamp, and DWT-DCT have significantly lower robustness, especially under stochastic regeneration attacks, their bit accuracy is generally lower than 0.5. This fully verifies that the VINE model has better generalization ability and cross-model stability, and its robustness advantage is obvious.

4. Prediction of the Robustness Potential of Low-Computation Large Model Editing Based on Low-Frequency Filtering

4.1. Motivation and Approach

In the original VINE paper, the authors identified a key observation: image editing performed by large generative models predominantly affects the low-frequency components of an image’s Gaussian spectrum, while the high-frequency regions remain comparatively stable. This finding directly motivates VINE’s design philosophy—embedding watermark information primarily in the high-frequency domain so that the watermark is naturally more resilient to edits generated by large models.

However, text-to-image models capable of high-quality image editing typically require substantial hardware resources and have very large model sizes. Directly evaluating the robustness of a watermarking model against such generative editing systems would therefore incur extremely high computational overhead. To address this issue, we design a low-cost evaluation method based on low-frequency filtering. By selectively attenuating the low-frequency components of an image in the frequency domain, this method approximates the distortions that large generative models impose on the information contained in the low-frequency spectral region, but at a fraction of the computational cost. In this way, we can preliminarily assess a watermarking model’s robustness against large-model edits without actually invoking heavy text-to-image systems. At the same time, we aim to use this tool to investigate the behavior of different watermarking algorithms along the frequency dimension and to provide a convenient screening and comparison mechanism for more complex attack scenarios, such as hybrid spatial–frequency or model-editing attacks.

4.2. Functional Extension under Low-Frequency Filtering Attacks

To extend VINE’s robustness from spatial distortions to frequency-domain perturbations, we introduce a differentiable low-frequency attenuation module into the training loop. This module is implemented as a high-pass filter in the FFT domain and is directly integrated into the encoder–decoder optimization process.

Through this integration:

- The network becomes aware of spectral corruption during backpropagation.
- The encoder learns to embed watermark signals into more stable high-frequency regions.
- The model gains improved adaptability to frequency-domain distortions.

This enhancement fundamentally expands VINE’s functional capacity. It is not merely an evaluation tool, but a model-level modification aimed at improving robustness under global spectral shifts.

4.3. Experiment Design

We systematically evaluate robustness under low-frequency filtering using three watermarking approaches:

- VINE (with functional extension)
- RivaGAN (deep watermarking baseline)
- DWT+DCT (traditional frequency-domain watermarking)

Dataset and Embedding:

A total of 100 images are selected from the STO_REGENERATION_1K subset of W-Bench-small.

All methods embed a fixed-length byte-sequence watermark into these images.

Frequency-Domain Attack Module:

A custom Java-based FFT pipeline performs low-frequency attenuation with eight cutoff values:

$D = \{1, 3, 5, 10, 20, 50, 100, 200\}$

Using the Gaussian high-pass filter:

$$H(u, v) = 1 - \exp\left(-\frac{D(u, v)^2}{2D_0^2}\right)$$

The module supports automated batch processing for all methods and all D values.

Evaluation Metrics:

For each attacked image:

- Bit-level accuracy: percentage of correctly recovered bits
- Image-level accuracy: 1 only if all bits are correct; otherwise 0

Averaging across 100 images yields performance for each D value.

D	VINE BIT ACCURACY	VINE PIC ACCURACY	RIVAGAN BIT ACCURACY	RIVAGAN PIC ACCURACY	DWT+DCT BIT ACCURACY	DWT+DCT PIC ACCURACY
1	0.98	0.42	0.95	0.48	0.76	0.00
3	0.99	0.53	0.92	0.14	0.82	0.00
5	0.99	0.46	0.87	0.02	0.86	0.00
10	0.99	0.48	0.80	0.00	0.91	0.00
20	0.99	0.44	0.71	0.00	0.93	0.00
50	0.95	0.11	0.52	0.00	0.96	0.02
100	0.91	0.03	0.41	0.00	0.94	0.00
200	0.87	0.01	0.41	0.00	0.90	0.00

Table 6: Accuracy results.

4.4. Results and Contributions

(1) Experimental Results

Low-frequency filtering produces distinct behavioral patterns across methods:

1. VINE demonstrates strong and stable robustness.

Across all D values, VINE maintains extremely high bit-level accuracy (often ≥ 0.95), confirming its ability to preserve watermark information even under severe spectral suppression.

Although image-level accuracy gradually declines with D , the degradation is smoother and substantially less severe than competing methods.

2. RivaGAN is highly vulnerable to spectral distortion.

Both bit-level and image-level accuracy drop sharply as D increases.

When $D \geq 5$, image-level accuracy approaches zero, indicating a lack of resilience to global frequency perturbations.

3. DWT+DCT shows local stability but fails globally.

Bit-level accuracy sometimes increases under stronger attacks due to redundancy in mid-frequency coefficients, but image-level accuracy remains nearly zero across all D values, suggesting weak global message integrity.

These findings collectively show that VINE maintains superior robustness under frequency-domain perturbations and degrades in a more controlled manner compared with both deep and traditional baselines.

(2) Methodological and Engineering Contributions

1. A lightweight surrogate for large-model editing robustness.

The proposed low-frequency attenuation method provides a low-cost approximation of spectral distortions caused by advanced editing models, reducing reliance on high-compute T2I systems.

2. A complete, reusable FFT-based attack module.

The Java implementation supports automated batch attacks across multiple distortion strengths, offering a unified testbed for a variety of watermarking methods.

3. Functional extension of VINE into the frequency domain.

This work expands VINE’s applicability from spatial editing robustness to frequency-domain perturbations, broadening its robustness profile.

4. A differentiable frequency-domain enhancement mechanism.

By integrating spectral corruption into training, VINE learns intrinsic frequency robustness—a capability absent in its original design.

5. An extensible evaluation framework for future research.

The complete pipeline—from unified embedding to multi-level accuracy assessment—serves as a foundation for studying hybrid spatial–frequency attacks.

(3) Overall Significance

This chapter enhances VINE both methodologically and functionally:

it introduces a practical framework for frequency-domain robustness evaluation, extends VINE’s applicability, and validates its superior stability under global spectral perturbations. The proposed approach provides a promising direction for future research into more complex and hybrid watermark attack scenarios.

5. Change the prior to the vine model to add new features

5.1. Motivation and Approach

The original VINE model adopts SD-Turbo as its diffusion prior. In terms of watermarking performance, this design is quite successful, but it has a serious practical drawback: SD-Turbo does not support LoRA or other parameter-efficient fine-tuning methods. As a result, any attempt to improve VINE essentially requires full-parameter fine-tuning, whose computational cost is close to pre-training a diffusion model from scratch. For a

research or course project, this makes the model almost impossible to iterate on.

In order to break this bottleneck, we decided to replace the prior with alternative diffusion backbones that offer comparable generation quality, and natively support LoRA-style fine-tuning. This change is not only a small engineering tweak, but a necessary step if VINE is to be treated as a practical, extensible watermarking framework rather than a fixed, “closed” model.

5.2. Experiment Design

We experimented with two SDXL-based models as replacements for SD-Turbo: Stable Diffusion XL Base 1.0 (SDXL-Base) and SDXL-Lightning. SDXL-Base emphasizes fidelity and stability, while SDXL-Lightning focuses on speed. Both are compatible with the LoRA ecosystem.

On the architectural level, we replaced the SD-Turbo components inside VINE with their SDXL counterparts, including the text encoder, UNet, and VAE modules. This required:

- aligning the latent space size and channel dimensions,
- adapting the scheduler configuration to match SDXL’s noise schedule,
- updating the embedding and secret-injection interfaces so that watermark bits can still be injected at the appropriate stages.

5.3. Contribution and Results

From an experimental perspective, the main value of this chapter lies not only in successfully training a new variant of VINE, but in making VINE practically improvable and extensible.

First, by replacing the SD-Turbo prior with SDXL-Base, we turn VINE from a model that effectively requires full-parameter fine-tuning into one that can leverage LoRA and other parameter-efficient techniques. This substantially reduces the entry barrier for future research: researchers no longer need extremely high-end GPUs to experiment with new loss functions, secret encoders, or task-specific adaptations. In other words, the model becomes experimenter-friendly rather than being locked behind hardware constraints.

Second, we demonstrate that the prior component of VINE is structurally interchangeable. The successful training of Vine-Base shows that, as long as the latent interface is matched, VINE can be combined with different diffusion backbones (e.g., SDXL-Base, SDXL-Lightning). This provides a general recipe for upgrading VINE in parallel with the diffusion model ecosystem, instead of treating the original SD-Turbo prior as a fixed, non-negotiable design.

Third, even under a relatively small dataset ($\approx 4,000$ images) and limited training steps, Vine-Base achieves

functional watermark embedding and decoding, with reasonable robustness in preliminary tests. This confirms that the proposed architectural and training changes are not merely theoretical but practically viable. Future work can reuse our implementation and focus on scaling up data and training time, rather than re-solving integration issues.

In short, the experiments in this chapter transform VINE from a static, hard-to-fine-tune model into a flexible framework that can evolve with new priors and lightweight training methods, which we believe is more valuable in the long run than a small gain in benchmark numbers.

5.4. Challenges

During the implementation, the main difficulty came from the high memory demand of SDXL-based models. Training became unstable under limited GPU resources, and version incompatibility also appeared when attempting to run on newer hardware. These issues were eventually resolved by switching to multi-GPU instances on AutoDL, adopting DeepSpeed for model parallelism, and performing iterative debugging to ensure the new architecture could be trained reliably.

6. Conclusion

This project establishes VINE as a practical, resilient, and extensible invisible watermarking framework suited for the challenges posed by modern generative AI editing. Through a redesigned encoder architecture, a Condition Adaptor for efficient feature fusion, and a progressive three-stage training strategy, we significantly enhance watermark extraction accuracy while preserving perceptual image quality. The introduction of W-Bench enables systematic stress-testing across a wide range of generative editing scenarios, offering clear evidence that VINE consistently outperforms classical signal-processing and baseline deep learning watermark models. Moreover, by replacing the SD-Turbo prior with SDXL-Base/Lightning, we equip VINE with LoRA-based fine-tuning capability, lowering computational barriers and making the framework more accessible for future research and real-world deployment. Additional functionality improvements—such as the frequency-domain attenuation evaluation and its differentiable counterpart integrated into training—expand VINE’s robustness beyond spatial distortions to spectral perturbations. Overall, the outcomes of this project demonstrate that VINE is not only a robust watermarking model but also a flexible platform capable of evolving with emerging generative technologies and supporting further research on watermark security.

7. Team Member Names and Responsibilities

Chi Yingshi: Maintain the project environment, design and conduct evaluation experiments, participate in improving model.

Huang Mingyu: Design and conduct evaluation experiments, process and analyze experiments data.

Zhu Qian: Search materials, write final report, make presentation slides, mainly responsible for organizing and coordinating the project.

Zhu Ruixiang: Design and develop the main part of model improvement, also conduct the majority of literature review.

8. Project Timeline with Milestones

Week1	Project initiation and literature review.
Week2-Week3	Environment configuration and model architecture implementation.
Week4-Week5	Attack simulation and robustness testing.
Week5-Week7	Model improvement and Low-Rank Adaptation (LoRA) adjustment.
Week6	Presentation preparation.
Week7	Report writing.

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